

constant and C is a function of disposal income which is also determined by Y and therefore, the S becomes constant which is denoted by $\bar{S}$.

Hence,

$$\bar{S} - I(r^*) = NX$$

The last equation clearly shows that net exports or trade balance depends upon the world interest rate r^* and $\bar{S}$.

Real Interest rate

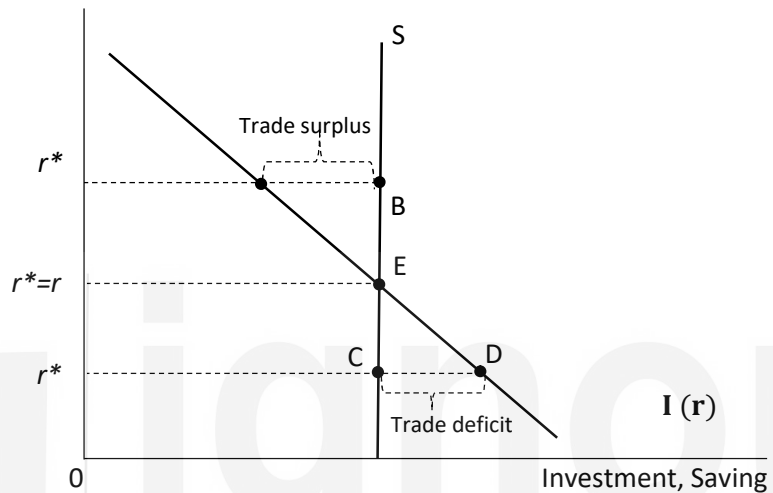


Fig. 28.3: Relationship between Real Interest Rate, Saving and Investment

Fig. 28.3 shows that with constant saving line and downward-sloping investment function determine the interest rate r if the economy is closed and it is in equilibrium at E . But when the economy is open and the world interest rate $r^* > r$, then there is a trade surplus equal to AB and when $r^* < r$, there is a trade deficit equal to CD . When $r^* = r$, then there is zero trade balance. Thus, in the case of a small open economy, the world interest rate determines the trade balance. Now, we see the relationship between the real exchange rate and the net export

We know that

Real Exchange Rate

$$= \text{Nominal Exchange Rate} \times \frac{\text{Price level in the domestic country}}{\text{price level in the foreign country}}$$

Let

$$\alpha = \text{Real exchange rate}$$

$$e = \text{Nominal exchange rate}$$

And

$$\theta = \frac{\text{Price level in the domestic country}}{\text{Price level in the foreign country}}$$

Therefore,

$$\alpha = e \times \theta$$

Now, for a given value of e , α will rise if θ rises or vice versa. Therefore, when the real exchange rate increases, then it means foreign goods have become relatively inexpensive and it will be beneficial to buy goods from abroad. As a result, imports from the home country will increase. At the same time, foreigners will not buy from the domestic country due to low relatively expensive goods as compared to their own country. As a result, the exports of the home country will decrease. In this way, the real exchange rate causes imports to increase and exports to fall. Overall, net exports fall. Hence, there is a negative relationship between the net exports and the real exchange rate. Mathematically, it can be said that net export is a negative function of the real exchange rate. Symbolically,

$$NX = f(\theta)$$

Graphically,

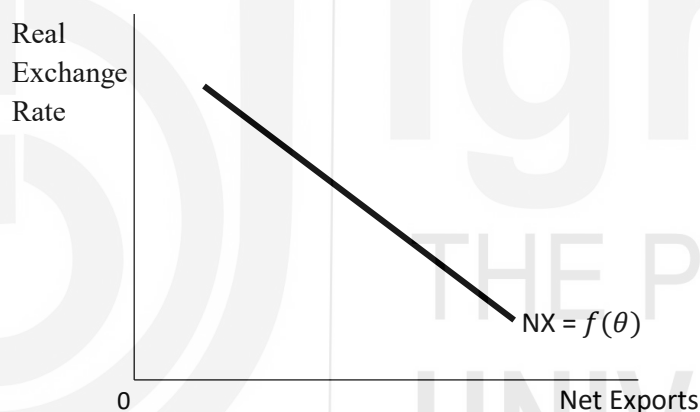


Fig. 28.4: Relationship between Real Interest Rate and Net Exports

Thus, the relative worth of currencies is shaped by the demand for them, which, in turn, is heavily influenced by trade dynamics. When a country consistently exports more than it imports, resulting in a trade surplus, there is a strong demand for its products and, consequently, its currency. According to the basic principles of supply and demand, high demand leads to higher prices and an increase in the value of the currency, which appreciates.

Conversely, when a country imports more than it exports, creating a trade deficit, the demand for its currency is relatively weaker. As a result, currency prices tend to decrease. In the context of currency, this means the currency depreciates or loses value.

6) Inflation and Interest Rates

Inflation and interest rates are closely intertwined, and their relationship can significantly affect exchange rates. Countries strive to strike a

delicate balance between these two factors, but managing this interplay is often intricate. Lower interest rates stimulate consumer spending and foster economic growth, typically exerting a positive impact on a country's currency value. However, if consumer spending surges to the extent that it outpaces supply, it can lead to inflation, which isn't necessarily a negative outcome. Nonetheless, lower interest rates typically do not attract foreign investment. On the other hand, higher interest rates tend to lure foreign investors, which, in turn, heightens the demand for a country's currency. Finding the right equilibrium in this regard can be challenging, and the effect on a country's exchange rate can be complex. In general, when inflation is high, it tends to weaken a currency as its purchasing power diminishes. This makes goods more expensive and can discourage investors. Conversely, when inflation is low, more capital flows into the country, bolstering the currency's value due to increased purchasing power and thus improving its exchange rate. In this regard, you should recall the IS-LM model in a closed economy and the **Mundell-Fleming Model** in an open economy.

7) Global Events

Events like financial crises, geopolitical tensions, and economic downturns in major economies can significantly impact exchange rates and disrupt stability. 2008 saw a severe decline in global currency values. The global financial crisis was a global issue that affected the Indian economy in a ripple fashion. After the Great Depression, this was the most serious crisis. Covid-19 caused an enormous crisis in 2020 that affected the entire planet. Every country's economy had difficulties as a result of the pandemic, including India. The value of the Indian rupee experienced fluctuation. The country did, however, recover more quickly than expected thanks to its fortitude and aggressive initiatives. Chortane and Pandey (2022)⁴ conducted an event study to find the impact of the Russia-Ukraine war on the value of global currencies against the US dollar. The study shows that the conflict between Russia and Ukraine affected the value of all major world currencies negatively, but a region-by-region analysis reveals that the currencies of the Pacific region significantly appreciated, the currencies of the Middle East and Africa (ME&A) are negligible, and European currencies (especially the Russian rouble, Czech koruna, and Polish zloty) depreciated against the USD. Further, the study shows how the currencies of Poland and the Czech Republic have depreciated considerably compared to the US dollar as a result of the financial and economic restrictions placed on Russia and their closeness to the conflict zone. Moreover, the Pan-American, European (especially the Polish zloty and the Russian rouble), and

⁴Chortane, S. G., & Pandey, D. K. (2022). Does the Russia-Ukraine war lead to currency asymmetries? A US dollar tale. *The Journal of Economic Asymmetries*, 26, Article e00265. <https://doi.org/10.1016/j.jeca.2022.e00265>

Middle Eastern and African (ME&A) currencies have all benefited greatly from the Russian Central Bank's announcement to tie the rouble with gold.

28.4 MEASURES TO MAINTAIN FOREIGN EXCHANGE STABILITY

Here are certain ways to maintain foreign exchange stability in an economy.

1) Central Bank Intervention

Central banks can use their foreign exchange reserves to buy or sell their currency in the foreign exchange market to influence its value. By doing so, they can mitigate excessive currency fluctuations and maintain stability. This is often done to counteract extreme currency appreciation or depreciation.

2) Fixed Exchange Rate Systems

Some countries opt for fixed or pegged exchange rate systems where the value of their currency is fixed or pegged to another stable currency, such as the U.S. dollar or the euro. Maintaining this peg requires active central bank intervention and foreign reserves to ensure that the exchange rate remains stable. However, the fixed exchange rate system has its limitations.

3) Managed Float

In a managed float system, exchange rates are allowed to fluctuate within a certain range, but central banks will intervene when the rate moves outside of that range to prevent extreme volatility.

4) Fiscal and Monetary Policies

Governments can use fiscal and monetary policies to influence exchange rates indirectly. For example, interest rate adjustments can affect the attractiveness of a currency to foreign investors. Tightening or loosening monetary policy can influence exchange rates.

5) Economic Reforms

Structural reforms in the economy, such as improving infrastructure, reducing trade barriers, and addressing fiscal deficits, can boost confidence in the currency and help maintain its stability.

6) Exchange Rate Bands

Implementing an exchange rate band system can allow for some degree of fluctuation while preventing extreme movements. The central bank intervenes when the exchange rate reaches the upper or lower bounds of the band.

7) **Transparency and Communication**

Central banks and governments should communicate their exchange rate policies clearly to the public and financial markets. Transparency can help manage expectations and reduce uncertainty.

8) **Foreign Exchange Reserves**

Accumulating and maintaining foreign exchange reserves provides a financial buffer that can be used to stabilize the currency during periods of volatility. However, building and maintaining sufficient reserves can be expensive.

9) **International Cooperation**

Collaboration with other countries and international organizations can help stabilize exchange rates. This can include currency swap agreements, coordinated interventions, and harmonized policies.

You should note that the approach to maintaining foreign exchange stability can vary from one country to another based on their economic conditions, policy objectives, and the global economic environment. Additionally, exchange rate stability is not always the primary goal; some countries may prioritize other economic objectives, such as inflation control or economic growth, which can affect exchange rates.

Check Your Progress 2

- 1) Explain the relationship between Real interest rate, Saving and Investment.
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- 2) Describe the relationship between real interest rate and net exports.
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- 3) State some of the latest issues of foreign exchange stability.
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- 4) What are your suggestions for maintaining foreign exchange stability?

28.5 LET US SUM UP

Foreign exchange stability refers to the ability of a country to maintain a relatively stable and predictable exchange rate for its currency in the foreign exchange market. In other words, it means that the value of a country's currency remains relatively constant over time, which can be important for economic and financial stability. Foreign exchange stability is essential for a country's economic well-being and international trade. Achieving and maintaining this stability involves a complex interplay of economic policies, market forces, and external factors. Countries strive to strike a balance that promotes growth, trade, and financial stability while avoiding excessive exchange rate volatility. Here are some key points related to foreign exchange stability and its related issues:

- a) **Exchange Rate Stability:** Exchange rate stability implies that the value of a country's currency remains steady or experiences only gradual fluctuations about other currencies. This stability can make it easier for businesses, investors, and governments to plan for the future.
- b) **Economic Factors:** Several economic factors can influence exchange rate stability, including inflation, interest rates, trade balances, and fiscal policies. A country with low inflation, stable interest rates, a trade surplus, and responsible fiscal policies is more likely to have a stable exchange rate.
- c) **Central Bank Interventions:** Central banks often play a crucial role in maintaining exchange rate stability. They can buy or sell their currency in the foreign exchange market to influence its value. For example, if a country's currency is appreciating too rapidly, the central bank may intervene by selling its currency to increase its supply and decrease its value.
- d) **Fixed vs. Floating Exchange Rates:** Countries can adopt different exchange rate systems. Some use fixed or pegged exchange rates, where the value of their currency is tied to a specific foreign currency (e.g., the US dollar). Others use floating exchange rates, which are determined by supply and demand in the foreign exchange market. Maintaining stability can be more challenging for countries with fixed exchange rates.
- e) **Speculation and Market Sentiment:** Exchange rates are also influenced by speculation and market sentiment. Traders and investors make decisions based on their expectations of future exchange rate movements, which can sometimes lead to sharp and unpredictable fluctuations.

- f) **External Shocks:** Events such as financial crises, geopolitical tensions including the war, or natural disasters can disrupt exchange rate stability. These external shocks may lead to sudden and significant currency depreciation or appreciation.
- g) **Currency Reserves:** Accumulating foreign exchange reserves can help a country maintain exchange rate stability. These reserves can be used to intervene in the foreign exchange market when necessary. However, building and maintaining sufficient reserves can be expensive.
- h) **International Cooperation:** Exchange rate stability is often a subject of international cooperation, especially within organizations like the International Monetary Fund (IMF). Countries may seek financial assistance or policy coordination to address exchange rate-related challenges.
- i) **Challenges of Overvaluation and Undervaluation:** An overvalued currency (one that is too strong) can hurt a country's exports, while an undervalued currency (one that is too weak) can lead to inflation and discourage foreign investment. Striking the right balance is essential for stability.

To maintain the foreign exchange stability, the following measures can be adopted.

- a) **Central Bank Intervention:** Central banks buy/sell currency to control value.
- b) **Fixed Exchange Rate:** Currency pegged to a stable currency, needs intervention.
- c) **Managed Float:** Exchange rates allowed to fluctuate within a range, intervened when extreme.
- d) **Forward Contracts/Options:** Used for hedging currency risk, and stabilizing rates.
- e) **Fiscal/Monetary Policies:** Adjusting interest rates influences currency attractiveness.
- f) **Economic Reforms:** Structural changes boost currency confidence.
- g) **Exchange Rate Bands:** Allow fluctuation within bounds, central bank intervenes at limits.
- h) **Transparency/Communication:** Clear policies reduce uncertainty.
- i) **Foreign Exchange Reserves:** Buffer for currency stabilization, costly to maintain.
- j) **International Cooperation:** Collaborative efforts for exchange rate stability.

28.6 KEY WORDS

Foreign exchange stability	: Consistency and predictability of exchange rates between different currencies.
Direct quote	: It measures how many units of domestic currency can be bought by one unit of a foreign currency.
Indirect quote	: It measures how many units of foreign currency can be bought by one unit of domestic currency.
Fixed Exchange Rate	: Under this regime, a country's currency is pegged to another currency or a fixed value like gold.
Floating Exchange Rate	: In this regime, the exchange rate is determined by market forces, such as supply and demand.
Managed float	: is a regime for exchange rates where the exchange rate is neither fully set nor completely free.

28.7 SOME USEFUL BOOKS

Case, K. E. (2012). *Principles of Macroeconomics*. Prentice Hall.

Krugman, P. R. (2012). *International Economics: Theory & Policy* (9th ed.). Pearson Addison-Wesley.

Mankiw, N. G. (2009). *Macroeconomics* (7th ed.). Worth Publishers.

Stutz, F. P., & Warf, B. (2010). *The World Economy: Geography, Business, Development*. Pearson Education, Limited.

28.8 ANSWER / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) It affects the cost of imports and exports, the competitiveness of industries, and the overall economic health of a country.
- 2) Receipts of exports, repayment of international loans, arbitrage purpose, speculation, gifts and grants from foreign countries, investments from foreign countries, and foreign remittances.
- 3) Refer to Sub-section 28.2.2

Check Your Progress 2

- 1) $\bar{S} - I(r^*) = NX$. Refer to section 28.3 for details.

- 2) Net export is a negative function of the real exchange rate. Symbolically, $NX = f(\theta)$
- 3) Managed or Dirty Floating, Exchange Rate Policy, Parallel Foreign Exchange Systems, Political Instability, Trade Balances, Inflation and Interest Rates, Global Events.
- 4) Central Bank Intervention, fixed Exchange Rate Systems, managed float, fiscal and monetary policies, economic reforms, exchange rate bands, transparency and communication, foreign exchange reserves, and international cooperation.

